§1 Symmetry

1. 经典力学 $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q}$

平移对称性：$L = \frac{1}{2}m\dot{x}^2$ $(x \to x+\varepsilon)$

↳ 动量守恒 $\frac{d}{dt}p_x = \frac{d}{dt}(m\dot{x}) = 0$

旋转对称性：$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) - U(r)$ $(\phi \to \phi + \varepsilon)$

↳ 角动量守恒 $\frac{d}{dt}p_\phi = \frac{d}{dt}(mr^2\dot{\phi}) = 0$

无穷小变换：

$$q_\alpha \to q_\alpha + \varepsilon\eta_\alpha$$

$$L(q_\alpha + \varepsilon\eta_\alpha,\ \dot{q}_\alpha + \varepsilon\dot{\eta}_\alpha) = L(q_\alpha, \dot{q}_\alpha) + \varepsilon\left(\eta_\alpha\frac{\partial L}{\partial q_\alpha} + \dot{\eta}_\alpha\frac{\partial L}{\partial \dot{q}_\alpha}\right)$$

$$= L(q_\alpha, \dot{q}_\alpha) + \varepsilon\left[\eta_\alpha\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) + \frac{\partial L}{\partial \dot{q}_\alpha}\frac{d}{dt}\eta_\alpha\right]$$

$$= L(q_\alpha, \dot{q}_\alpha) + \varepsilon\frac{d}{dt}\left(\eta_\alpha\cdot\frac{\partial L}{\partial \dot{q}_\alpha}\right) = L(q_\alpha, \dot{q}_\alpha) + \varepsilon\frac{d}{dt}(\eta_\alpha p_\alpha)$$

Symmetry：$\delta L = 0$

⇓

Conservation：$\frac{d}{dt}\sum_\alpha \eta_\alpha p_\alpha = 0$

e.g. 二维平面运动

$$L = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) - U(\sqrt{x^2+y^2})$$

Rotation about z：$(x, y) \to (x - \varepsilon y,\ y + \varepsilon x)$

$$\delta L = \frac{1}{2}m\left[(\dot{x}-\varepsilon\dot{y})^2 - \dot{x}^2 + (\dot{y}+\varepsilon\dot{x})^2 - \dot{y}^2\right] - U\left(\sqrt{(x-\varepsilon y)^2+(y+\varepsilon x)^2}\right) + U(\sqrt{x^2+y^2})$$

$$= \frac{1}{2}m\left[\cancel{-2\varepsilon\dot{x}\dot{y}} + \varepsilon^2(\dot{x}^2+\dot{y}^2) + \cancel{2\varepsilon\dot{y}\dot{x}} + \varepsilon^2(\dot{x}^2+\dot{y}^2)\right] - U\left[\sqrt{(1+\varepsilon^2)(x^2+y^2)}\right] + U(\sqrt{x^2+y^2})$$

$$= O(\varepsilon^2)$$

$$\to \frac{d}{dt}\left(p_x\varepsilon y + p_y\varepsilon x\right) = \frac{d}{dt}L_z = 0$$

2. Hamilton Mechanics

考虑一个无穷小正则变换

$q \to Q = q + \varepsilon\eta$, $p \to P = p + \varepsilon\xi$

$[Q,P] = [q+\varepsilon\eta,\ p+\varepsilon\xi]$

$= [q,p] + \varepsilon([\eta,p] + [q,\xi]) + O(\varepsilon^2)$

$= [q,p] + \varepsilon\left(\frac{\partial\eta}{\partial q} + \frac{\partial\xi}{\partial p}\right) = [q,p]$ 正则变换保持 $[q,p]$ 不变。

$\Leftrightarrow \frac{\partial\eta}{\partial q} + \frac{\partial\xi}{\partial p} = 0$

$\exists\ G(q,p)$, s.t. $\eta = \frac{\partial G}{\partial p}$, $\xi = -\frac{\partial G}{\partial q}$

$\Rightarrow Q = q + \varepsilon\frac{\partial G}{\partial p} = q + \varepsilon[q,G]$

$P = p + \varepsilon[p,G]$

对一个物理量 $f(q,p)$

$f \to f + \varepsilon[f,G]$

G 称为变换的生成元.

$[H,f] = 0 \Leftrightarrow$ ① H 在 f 生成变换下不变 $H = H + \varepsilon[H,f]$

② f 随时间守恒 $\frac{df}{dt} = \frac{\partial f}{\partial q}\frac{dq}{dt} + \frac{\partial f}{\partial p}\frac{dp}{dt} = \frac{\partial f}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial H}{\partial q}$

$= [f,H] = 0$

3. Liouville Integrability. Why Conservation Quantity Important?

H-J Eq. $p = \frac{\partial S}{\partial q}$

$H(q, \partial_q S, t) + \partial_t S = 0$ $\quad S = \int_{t_0}^{t} L(q, \dot{q}, t)\, dt$.

关于 $S(q,t)$ 的一阶微分方程.

Liouville - Arnold 可积定理:

时间由度为 N 的系统，如果有 N 个守恒量 $I_i(q,p,t)$

且 $\forall i,j,\ \{I_i, I_j\} = 0$

则 $dS = \sum_i p(q, I, t)\, dq - H\{q, p(q, I, t)\}\, dt$

考虑上 n 个守恒量 $\Rightarrow$ 对 $(q,p) \in T^n = \prod_i S^1$

定义作用量-角变量 $\{I_i; \theta_i\}$: $\theta_i = \frac{\partial H}{\partial I_i}$

$\hookrightarrow \theta_i = \theta_{i0} + \omega_i t$

4. QM Sym.

对对称性 U 保持内积模长 (既定概率) $|\langle U\phi | U\psi\rangle| = |\langle \phi | \psi\rangle|$ ① ②

Wigner Theorem : $U \to$ Unitary / Anti-Unitary.

① U : $\langle U\phi | U\psi\rangle = \langle \phi | \psi\rangle = \langle U^\dagger U\phi | \psi\rangle$, $U^\dagger U = I$

② $\forall U$: $\langle U'\phi | U'\psi\rangle = \langle \phi | \psi\rangle^* = \langle U^\dagger U\phi | \psi\rangle^*$, $U'^\dagger U' = I$

Prove : $|i\rangle \to |\vec{i}\rangle = \hat{U}|i\rangle$, $\langle \vec{i} | \vec{j}\rangle = \delta_{ij}$

$|\psi\rangle = \sum_i c_i |i\rangle$, $|\vec{\psi}\rangle = \sum_i \vec{c}_i |\vec{i}\rangle$

Let $|\phi\rangle =$

$|i\rangle$ $\quad |c_i| = |\vec{c}_i|$ ①

$\frac{1}{\sqrt{2}}(|i\rangle + |j\rangle)$ $\quad |c_i + c_j| = |\vec{c}_i + \vec{c}_j|$ ②

$\frac{1}{\sqrt{3}}(|i\rangle + |j\rangle + |k\rangle)$ $\quad |c_i + c_j + c_k| = |\vec{c}_i + \vec{c}_j + \vec{c}_k|$ ③

① $\to c_i = r_i e^{i\alpha_i}$, $\vec{c}_i = r_i e^{i\vec{\alpha}_i}$ 有上简式.

② : $\mathrm{Re}(c_i^* c_j) = \mathrm{Re}(\vec{c}_i^* \vec{c}_j) \Rightarrow \cos(\alpha_i - \alpha_j) = \cos(\vec{\alpha}_i - \vec{\alpha}_j)$

③ : $\cos\Delta\alpha_{ijk} = \cos\Delta\vec{\alpha}_{ijk}$

$\Rightarrow \alpha \cdot \vec{\alpha} = \alpha^2$ $\quad (\alpha)$

b. $\hat{\alpha} = -\alpha\ |A|\psi\rangle$

3. Sym. → Good State.

if $[A, H]$

$$\begin{cases} H(A|\psi_n\rangle) = AH|\psi_n\rangle = E_n(A|\psi_n\rangle) \\ A(H|\psi_a\rangle) = HA|\psi_a\rangle = a(H|\psi_a\rangle) \end{cases}$$

A, H有共同本征态, a: 数量之数.

6. In QM: $[A, B] \to \frac{1}{i\hbar}[\hat{A}, \hat{B}]$

生成元 $\hat{G}$ ~ 无穷小变换 $\hat{u} = (1 - \frac{i}{\hbar}\varepsilon\hat{G})$ 这里 u 为酉算子之为的无法 $\lim_{\varepsilon\to 0} u_\varepsilon = I$.

$$\hat{u}^\dagger\hat{u} = I = I + \frac{i\varepsilon}{\hbar}(\hat{G}^\dagger - \hat{G})$$

$$u(\theta) = \lim_{N\to\infty}(1 - \frac{i}{\hbar}\frac{\theta}{N}\hat{G})^N = e^{-\frac{i\theta\hat{G}}{\hbar}}$$

$$\hat{f}' = \hat{u}^\dagger\hat{f}\hat{u} = \exp(\frac{i}{\hbar}\varepsilon\hat{G})\hat{f}\exp(-\frac{i}{\hbar}\varepsilon\hat{G})$$

$$= (1 + \frac{i}{\hbar}\varepsilon\hat{G})\hat{f}(1 - \frac{i}{\hbar}\varepsilon\hat{G})$$

$$= \hat{f} + \frac{1}{i\hbar}[\hat{f}, \hat{G}]\cdot\varepsilon + O(\varepsilon^2)$$

$$\therefore \delta\hat{f} = \frac{\varepsilon}{i\hbar}[\hat{f}, \hat{G}]$$

e.g. Translation

$$\hat{u}^\dagger(a)\,\hat{x}\,\hat{u}(a) = \hat{x} + \frac{\varepsilon}{i\hbar}[x, \hat{G}] = \hat{x} + \varepsilon$$

$$\to \hat{G} = \hat{p},\ \hat{u}(a) = \exp(-\frac{ia\hat{p}}{\hbar})$$

$$\hat{u}(a)|x\rangle = |\hat{x}+a\rangle$$

$$\langle x|\hat{u}(a)|\psi\rangle = \langle\hat{u}^\dagger(a)x|\psi\rangle = \psi(x-a)$$

e.g. 2 Rotation about z

$$\hat{u}^\dagger(\varepsilon)\,\hat{\phi}\,\hat{u}(\varepsilon) = \hat{\phi} + \frac{\varepsilon}{i\hbar}[\hat{\phi}, \hat{G}] = \hat{\phi} + \varepsilon$$

$$\to \hat{G} = \hat{L}_z,\ \hat{u}(\theta) = \exp(-\frac{i\theta\hat{L}_z}{\hbar})$$

$$\hat{u}(\theta)|\phi\rangle = |\phi+\theta\rangle$$

$$\langle\phi|\hat{u}(\theta)|\psi\rangle = \psi(\phi-\theta)$$

eg. 3. Scale Transformation

$$\hat{U}(a)|x\rangle = e^{-\frac{ia}{\hbar}\hat{p}}|x\rangle = |x+a\rangle$$

$$\langle x|\hat{U}(a)|\psi\rangle = \psi(x-a)$$

$$\hat{U}(a) = e^{-\frac{ia}{\hbar}\hat{p}} = 1 - \frac{ia}{\hbar}\hat{p} + \frac{1}{2!}\left(-\frac{ia}{\hbar}\right)^2\hat{p}^2 + \cdots$$

$$\hat{p} = \frac{\hbar}{i}\frac{d}{dx}$$

$$\hat{U}(a)\psi(x) = \psi(x-a) = \psi(x) - a\frac{d\psi}{dx} + \frac{1}{2!}a^2\frac{d^2\psi}{dx^2} + \cdots$$

$$= \left(1 - a\frac{d}{dx} + \frac{1}{2!}a^2\frac{d^2}{dx^2} + \cdots\right)\psi(x)$$

$$= e^{-a\frac{d}{dx}}\psi(x)$$

$$\hat{U}(a) = e^{-a\frac{d}{dx}}$$

7. Multi-Generator

$$\hat{J}_z \sim U_z(\varepsilon) = \exp\left(-\frac{i}{\hbar}\varepsilon\hat{J}_z\right); \quad \hat{Q} \sim U_Q(\varepsilon) = \exp\left(\frac{i}{\hbar}\varepsilon\hat{Q}\right) dx$$

$$\begin{cases} \hat{J}_z\hat{Q} - \hat{Q}\hat{J}_z = i\hbar\hat{Q} \\ \hat{Q}^2 - \hat{P}^2 = \frac{i}{\hbar}(\hat{Q}\hat{P} + \hat{P}\hat{Q}) \end{cases}$$

$$U_z(\varepsilon)U_Q(\varphi) = \left(1 - \frac{i}{\hbar}\varepsilon\hat{J}_z\right)\left(1 - \frac{i}{\hbar}\varphi\hat{Q}\right)$$

$$= 1 - \frac{i}{\hbar}\varepsilon\hat{J}_z - \frac{i}{\hbar}\varphi\hat{Q} - \frac{1}{\hbar^2}\varepsilon\varphi\hat{J}_z\hat{Q}$$

$$= 1 - \frac{i}{\hbar}\varphi\hat{Q} - \frac{i}{\hbar}\varepsilon\hat{J}_z\hat{Q} + \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 - \frac{i}{\hbar}\varphi\hat{Q} - \frac{i}{\hbar}\varepsilon\left(\hat{Q}\hat{J}_z - \hat{J}_z\hat{Q}\right) + \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 - \frac{i}{\hbar}\varphi\hat{Q} - \frac{i}{\hbar}\varepsilon\left(i\hbar\hat{Q}\right) + \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 + \varepsilon\hat{Q} + \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 + \varepsilon\hat{Q} + \frac{1}{\hbar^2}\varepsilon\varphi\left(i\hbar\hat{Q} + \hat{Q}\hat{J}_z\right)$$

$$= 1 + \varepsilon\hat{Q} + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} - \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} - \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} - \frac{1}{\hbar^2}\varepsilon\varphi\left(i\hbar\hat{Q} + \hat{Q}\hat{J}_z\right)$$

$$= 1 + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} + \frac{1}{\hbar}\varepsilon\varphi\hat{Q} - \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} - \frac{1}{\hbar^2}\varepsilon\varphi\hat{Q}\hat{J}_z$$

$$= 1 + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} + \frac{i}{\hbar}\varepsilon\varphi\hat{Q} +

$A \to D^{\dagger}(g) A D(g)$

$\psi(x) = \langle x|\psi\rangle \to \langle x|D(g)|\psi\rangle = \langle D^{\dagger}(g)x|\psi\rangle = \langle g^{-1}x|\psi\rangle = \psi(g^{-1}x)$

9. Rotation in 3D space

$v \in \mathbb{R}^3$, $v \mapsto Rv$

长度保持长度不变：$v^T R^T R v = v^T v \to R^T R = I$.

无穷小转动：$R(w) = I + w$

$(I+w)^T(I+w) = w^T + w = 0$

令 $w = \begin{pmatrix} 0 & -\varepsilon_3 & \varepsilon_2 \\ \varepsilon_3 & 0 & -\varepsilon_1 \\ -\varepsilon_2 & \varepsilon_1 & 0 \end{pmatrix}$ $L_1 = t_0\begin{pmatrix} & & \\ & & -i \\ & i & \end{pmatrix}$, $L_2 = t_0\begin{pmatrix} & & i \\ & & \\ -i & & \end{pmatrix}$, $L_3 = t_0\begin{pmatrix} & -i & \\ i & & \\ & & \end{pmatrix}$

$\Rightarrow R(w) = I - \frac{i}{t_0}\sum_i \varepsilon_i L_i$

$[L_i, L_j] = i t_0 \varepsilon_{ijk} L_k$, so(3) algebra.

In Hilbert space:

R's representation $U(R)$:

$\vec{r} \to \vec{r}' = \hat{R}\vec{r}$, while $|\psi\rangle \to |\psi'\rangle = U(R)|\psi\rangle$, $\vec{r}' = \langle\psi|\hat{\vec{r}}|\psi\rangle$

$\hat{R}\hat{\vec{r}} = \langle\psi'|\hat{\vec{r}}|\psi'\rangle = \langle\psi|U^{\dagger}(R)\hat{\vec{r}}\,U(R)|\psi\rangle$

$\Rightarrow U^{\dagger}(R)\hat{r}_i U(R) = \sum_j R_{ij}\hat{r}_j$

$\Rightarrow$ $U(R)$ 的生成元 J ~ R 的生成元（角动量）L

$U(R_\varepsilon) = 1 - i\varepsilon\frac{J}{t_0}$, $R_\varepsilon = 1 - i\varepsilon\frac{L}{t_0}$

$[J_i, J_j] = i t_0 \varepsilon_{ijk} J_k$

① $\vec{J} = \vec{r}\times\vec{p}$

② $\vec{J} = \frac{t_0}{2}\vec{\sigma}$ $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$